



GREEN TECHNOLOGY

IS INDIA RENEWABLE ENERGISED?

"Earth provides enough to satisfy every man's needs, but not every man's greed"—Mahatma Gandhi

A recent report published by Niti Aayog highlights that India's current installed electricity generation capacity of 275GW is significantly higher than the peak demand of 140GW. The capacity level of coal, which dominates as the primary energy source in India, is higher than its peak demand. Then, why is the country facing power shortage? Reasons are plenty, including:

1. High transmission and distribution losses at the distribution company (discom) level
2. Sub-standard health of infrastructure units supporting power generation facilities
3. Shortage of coal supply across all generation points due to connectivity issues
4. Unregulated price of imported coal

Owing to these and other pertinent issues, discoms, which purchase energy from these plaguing power generation units, face the problem of higher input costs, which ultimately affects the consumer! So when you face a power outage in your home today, don't think that energy production is low - maybe the discom that supplies power to your home has refrained from buying adequate power from generation units in order to cut facility cost.

That's the reason why our honourable Prime Minister Shri Narendra Modi talks about the seven horses of energy. He emphasises on evolution beyond the plaguing areas to newer sources of energy

generation like solar, wind and biomass (refer Fig. 1 for details). Today, India has more than 300 million people who don't have electricity in their homes, and are still waiting for the first glow of light in their homes. In order to provide electricity to their houses, the country just can't rely on traditional methods of power generation, and would need to look at expanding base towards renewable energy (RE).

In my recent trip to the Hindi heartland of India, I was confronted with situations which I had never seen while growing up in a metropolis. Power cuts lasting more than ten hours were just like a norm. As a solution, people resort to costly backup solutions like diesel generators, which pose great risks to both human health and environment. And to top it all, estimates suggest that by 2021-22 India's electricity demand will be more than double the level in 2011-12. So how to solve this quagmire?

The answer lies in exploring renewable energy resources, which are vast and largely unexplored. Forecasts show that India's solar energy potential is greater than 750GW, while installable wind energy potential is 302GW (actual figure could be higher than 1000GW). The India Energy Security Scenarios 2047 report launched in 2015 shows the possibility of achieving 410GW of wind energy and 479GW of solar PV energy by 2047. The potential of biomass and small hydro projects is also significant. Thus, renewable energy has the potential to anchor the development of India's electricity sector and reach the hinterlands not even thought of.

Now the question before the industry is "how do we reach economies of scale both by and through RE?" The government can provide the broad policy parameters and frameworks, but it's the industry which needs to act and support these avenues towards a conclusive

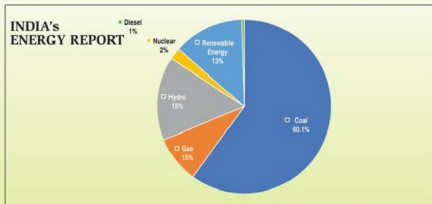


Fig. 1

Source : Ministry to New and Renewable Energy

